

ASSIGNMENTS

1. What type of data is stored in a FASTA file?

A FASTA file is the simplest way to store biological sequences. It stores raw text representing either nucleotide sequences (DNA/RNA) or amino acid sequences (proteins).

2. Why do we use BAM instead of SAM?

SAM (Sequence Alignment Map) and BAM (Binary Alignment Map). SAM is human- readable and is a large text file that can be opened and read in a terminal but the file size is massive.

While

BAM is machine- readable and is a compressed binary version of SAM .

BAM is much smaller (saving storing space) and faster for computers to process using indexing.

3. What do REF and ALT mean in a VCF file?

A VCF(Variant call format) file is used to record the difference between a sample and a known standard.

- REF(Reference): This is the standard nucleotide base found at a specific position in the reference genome (e.g., G).

- ALT (Alternate): This is the mutated or different base found in a specific sample at that same position(e.g., an A).